

Executive summary — Visual Augmentation of Audio Semantics for Accessibility

For someone stepping into this project today (updated 2026-09-15, night). Plain language; every technical word is explained the first time it appears, and the recurring ones are collected in the glossary at the end (§13). Every number here comes from a file in the repository. The full record is in `docs/project_notes.tex` (the dated lab notebook), `docs/report/report.pdf` (the thesis draft, 16 pages) and `LIMITATIONS.md` (what the numbers do not support).

1. The problem, in one minute

A deaf viewer gets a video's speech through subtitles, but not the rest of the soundtrack: a siren behind the camera, a dog barking in the next room, glass breaking off screen. Subtitle guidelines even tell captioners to *skip* sounds the picture already shows, and in practice most ambient sound is never captioned at all.

This project built a system that listens to a video, and for each ambient sound decides whether the thing making it is **already visible on screen**. If it is not, the system shows a small picture of that sound next to the video for as long as the sound lasts. If it is (you can see the dog barking), it shows nothing. That "show only when not visible" rule is called the **gate**. It is the whole idea, and the thing the evaluation tests.

Rules set by the supervisor: **no training** — the system is built by connecting existing, freely available AI models; **minimum human involvement** — decisions are made by models, not by hand-written rules; finish within the week.

2. The headline result, in three sentences

On 100 test clips, scored 0–4 by an automatic judge, our gated system scores **3.17** and the simplest alternative — "draw a picture for every sound you hear, never look at the video" (the **blind** baseline) — scores **3.16**: a tie. We win on the 50 clips where no picture was needed (we stay quiet, blind draws something redundant) and lose on the 50 where one was needed (we sometimes stay quiet when we should not). The reason we lose is mostly not the gate but the **sound detector** underneath it, and six attempts to improve that detector, each with a pass/fail rule written down beforehand, all failed — which is itself the thesis's

second finding.

3. The story of the project (what happened, in order)

- **Early August.** Project pivoted to its current form: pictures for heard-but-not-seen sounds. Proposal written with a seven-stage pipeline and an automatic evaluation.
- **Up to 8 Sept.** First working prototype: a sound detector (PANNs), a vision-language model to decide visibility, an image generator (SDXL), a side panel. Adam labelled a 274-clip benchmark. Many hand fixes made on one clip broke on the next (lessons in §7).
- **8–9 Sept.** Better object finder (OWLv2); evaluation split into "describe" and "judge" passes; FLUX replaces SDXL for pictures; an experiment showed a *stock photo* beats a generated picture by +0.17, but 55% of the photos had restrictive licences, so generation stayed.
- **12 Sept.** BEATs replaces PANNs as the detector (fixed 11 of 14 logged complaints).
- **13 Sept.** Process rules adopted: clips split into a development set (for choosing settings) and a test set (for the final score); every failure logged; a pass bar declared before every experiment. First outside check on the DCASE dataset (visibility decision 66.7% right). Version **v3** of the system submitted to the cluster (10 h).
- **14 Sept.** The previous version (**v2**) loses to blind under every way of scoring (−0.70 at worst). Found that the proposal's reference sentences were written by the system's own models — circular. Built a fully independent reference; found it is at chance (55%) on the one question that matters, so it cannot score the gate. **v3 lands: a tie (+0.01)**. A sweep of 16 gate settings on the development clips shows the current one is already best. Thesis draft written.
- **15 Sept.** Six detector improvements tried, all failed their declared bars (§10). A second, judge-free way of scoring (cost-sensitivity, §9) shows the gate is preferred once a redundant picture is counted as $\geq 40\%$ as bad as a missing one. Two things the judge cannot see were found (§8). Figures, bibliography, and this summary finished.

4. Background: what already exists

- **Subtitles** carry speech well and ambient sound badly. **Sign language** is richer but rare and cannot be automated.

- **Sound-awareness apps for deaf users** (HomeSound, SoundWatch) recognise ~20 everyday sounds in the home — a knock, an alarm, a crying baby — and users prefer pictures to text. They are for real life, not for watching video.
- **Sound detectors:** AI models trained on *AudioSet* (a Google collection of 2 million YouTube clips labelled with 527 kinds of sound) can name what is in a soundtrack. We use one called **BEATs**.
- **Vision-language models (VLMs):** AI models you can show pictures to and ask questions in words ("what is making this sound?"). We use **Qwen2.5-VL-7B** (mid-size; runs on one GPU).
- **Audio-to-image systems** (e.g. Sound2Scene) turn a sound directly into a picture, without looking at the video. They are our main comparison: "draw everything you hear".
- We found no existing system that does all three: detect ambient sound → decide whether it is heard-but-not-seen → show a picture. That combination, plus a way to test it, is the contribution.

5. How the system works (seven stages, code in `src/`)

stage	what it does	model
1	pulls the audio out of the video	FFmpeg
2	quick list of objects seen anywhere in the clip (stage 5 makes the real decision)	OWLv2
3	writes down any speech — used only as a hint, never drawn	Whisper
4	detects ambient sounds and when they start and stop	BEATs
5	reasons: is the source visible? what should the picture show? which sounds are one source?	Qwen2.5-VL-7B
6	draws the picture and places it beside the video	FLUX.1 (an image generator)
7	evaluates the result	see §8

The important behaviours (all decided by models at run time; settings in `config.py`)

Detection (stage 4). BEATs looks at the audio two seconds at a time, moving forward a quarter of a second per step, and gives each of the 527 sound kinds a

confidence between 0 and 1. A sound counts if its confidence reaches **0.35** (the "bar"). Two timing refinements: the start is stamped where the confidence *begins to rise*, not where it crosses the bar ("hysteresis"), so the picture is not late — this cut timing error from 0.43 s to 0.31 s on an outside dataset; and a further "occlusion" step that made no measurable difference was kept because v3 was scored with it on. Speech and music are never shown.

The visibility decision (stage 5, the gate). For each detected sound:

1. Its duration is cut into pieces of at most 5 seconds (a "stretch"). For each stretch, 6 video frames are taken, from 1 s before to 1 s after it.
2. The VLM is asked three differently worded questions about those frames (three "votes"):
 - *Name it*: "What in these frames is making the sound of X, or nothing?" If it names something, a follow-up checks common sense: "Does a make an X sound?" — so a cowboy hat cannot be the source of a vehicle sound.
 - *Is it happening*: "Can you SEE X happening — the source in frame and visibly making that sound?" Two options, asked twice with the options swapped; counted only if both answers agree (the model otherwise tends to pick the second option whatever it says).
 - *Describe*: "Describe what is in these frames." If the description names the source, or common sense says what it describes makes that sound, that counts as a vote.
3. Majority of the three votes = "visible" for that stretch. The sound is silenced only if it is visible in **every** stretch; otherwise the picture stays for the whole sound (one sound, one continuous picture — no flicker).

"Visible" means *seen making the sound*: a baby held in its mother's arms is not visibly crying; a fire-alarm box on a wall is not visibly ringing; a woman with her head back is visibly laughing.

What the picture shows. The VLM writes a 2–5-word phrase describing the *event* — "glass shattering", "audience clapping" — from the detector's label, the frames (a window or a bottle?) and a short place name ("kitchen"). Dialogue is never used (it once leaked into pictures). A second question checks the phrase points back to the right sound. FLUX draws it on a white background; the panel holds at most three pictures at once, each for at least 1.5 s.

One picture per sound source ("dedup"). The detector often gives one sound several names (Laughter, Giggle, Snicker). AudioSet's own family tree (Giggle is a kind of Laughter) merges them; a text-similarity check catches paraphrases the tree cannot; and when two names have exactly the same start and end (a crying baby heard as "Sheep" and "Baby cry"), the VLM looks at the frames and picks

one.

Speech as a hint. If people on the soundtrack react to a sound ("whoa, what's that?"), that sound gets priority when the panel is full. It never overrides the visibility decision.

6. One clip through all seven stages

Test clip `ly_ambulance_(siren)_-yPSgCn.mp4` (18 s): a street filmed from a car; an ambulance siren is heard but no ambulance is ever in frame. This is example (c) in the report's Figure 2.

1. **Audio out** (FFmpeg): an 18-second wav.
2. **Objects seen** (OWLv2): road, van, cars, train — a list for later.
3. **Speech** (Whisper): none.
4. **Detect** (BEATs): "Siren" confidence rises above 0.35 almost at once and stays there — peak 0.84, span 0.0–17.8 s. "Vehicle" is also heard (peak 0.61).
5. **Reason** (Qwen2.5-VL, for "Siren"): the 18 s become four stretches. In every one, *Name it* answers "nothing", *Is it happening* answers "no", and only *Describe* counts as a yes (the description mentions vehicles that could carry a siren). One vote of three → not visible, in all four stretches → **show a picture for the whole sound**. Phrase written: "ambulance siren blaring". ("Vehicle" was visible in three stretches but not the fourth, so it also gets a picture.)
6. **Draw** (FLUX): an ambulance with lights on, white background, in the side panel for the whole clip.
7. **Score** (see §8): the reference says a hearing viewer gets "the siren and vehicle sounds" that a deaf viewer would miss. A VLM looks at our panel and writes "an ambulance with its lights and siren activated"; the judge gives **4**. Blind also drew pictures but its VLM description talked about brake lights and traffic, not a siren: **3**. The text-caption baseline ("The soundtrack contains: Siren, Vehicle") got **2**.

7. Lessons from building it

Every early fix was made on the clip in front of us and broke on the next one. The rules that came out of that:

1. **Lists fail.** A hand-written list of 30 "visible objects" could never mark laughter as visible. Every list became a question to a model.
2. **Ask for a name, not a yes.** A yes/no question to a VLM gets agreement, not

evidence. A name can be checked.

3. **Read what a model already gives you.** Deduplication was solved by AudioSet's family tree after three cleverer ideas failed.
4. **Set thresholds once, on separate clips.** Since 2026-09-13: settings are chosen on a *development split* (clips not used for the final score), every failure is logged (`docs/failure_catalogue.md`), and a change needs three cases of one pattern.
5. **Evaluation code produces plausible wrong numbers.** An empty result scored 2/4; a similarity returned 0.0 for everything after a library update; an audio model was never given its audio and answered "none" 100 times. Each was found by reading logs.
6. **Declare the pass bar before running.** Every experiment this week wrote its pass/fail rule into the code before the job was submitted (see §10).

8. How a clip is scored (and why the score is lopsided)

There is no user study; scoring is automatic (as the proposal specified). For each clip and each system:

1. The system produces its output — pictures beside the video, or nothing.
2. A VLM looks at the result and writes what a viewer would learn from the pictures.
3. A **reference sentence** states what a hearing viewer gets from the soundtrack that a deaf viewer would miss — or "nothing beyond the picture" if nothing is missing.
4. A **judge** (a text model, Mistral-7B) scores how well (2) matches (3), from 0 to 4. Two cases are decided by code: showing nothing when something was missing = **0**; showing nothing when nothing was missing = **4**.

Two baselines are scored the same way: **blind** — draw a picture for every sound heard, never look at the video; **caption** — write the sound's name as text instead of a picture.

Two real clips show the lopsidedness.

- *Police car on screen* (`un_police_car_siren_WEROVWDp.mp4`, Figure 2a): the siren's source is visible, so the reference says "nothing beyond the picture". We stay silent → **4**. Blind draws a police car anyway → **3**. Being right here gains us **1 point**.

- *New York street* (`ambient_citywalk_nyc_2627.mp4`, Figure 2b): cars are visible, but the labelled sound is off-screen traffic, so the reference says a picture is due. Our gate saw the cars and stayed silent → **0**. Blind drew traffic → **3**. Being wrong here costs us **3–4 points**.

So a wrong silence costs about four times what a redundant picture costs. The gate must be right about 80% of the time when it stays silent just to break even. This shapes every result below. Whether deaf viewers feel the same 4-to-1 is unknown (§9 looks at other rates).

Who writes the reference sentence matters. Three versions:

- *model-derived* (the proposal's): written by the system's own models from its own detections. Found to be circular — it rewards whichever system repeats the detector. On the police-car clip above, this reference said the siren was *missing* (because the detector heard one) and gave our correct silence a **0**.
- *human-grounded*: the same, corrected by Adam's labels (on seen / no-ambient clips it becomes "nothing beyond the picture"). **This is the one the headline uses**. On the police-car clip it gives the correct silence its **4**.
- *independent*: written by a different set of models that never saw our system. It fixes the circularity, but its own guess at "is the source visible?" is no better than a coin flip (55% agreement with the labels), so it cannot score the gate. Reported, not used.

Two things the judge cannot see (found 2026-09-15): if the detector hears *nothing* on a clip where a picture was due, the reference itself says nothing is missing and *every* system gets 4 — so a detector miss is invisible to the score. And the score cannot say whether a picture was shown at the right moment; only that it was shown.

How reliable is the judge? A second, different judge model agreed with Mistral on 97% of clips to within one point and reproduced the gated-vs-blind gap — but it moved the caption baseline by +0.61, so any ranking involving the text baseline depends on which judge you ask.

9. Results (v3 = the current system; 100 test clips; score 0–4)

reference	gated (ours)	blind	caption	ours – blind
model-derived	2.64	3.04	2.79	–0.40 (clearly behind)

human-grounded	3.17	3.16	2.89	+0.01 — a tie (21 clips won, 64 tied, 15 lost)
----------------	------	------	------	--

The tie's 95% range is -0.15 to $+0.18$: the data cannot tell the two apart. Any mean over 25 clips carries roughly ± 0.2 , so small differences below are indications, not proof.

Split by whether a picture was due (human-grounded):

- **no picture due** (50 clips): ours 3.56 vs blind 3.30 → **+0.26, we win** — we correctly stay quiet, blind draws a redundant picture.
- **picture due** (50 clips): ours 2.78 vs blind 3.02 → **-0.24, we lose** — sometimes we stay quiet when we should not.

In words: *the system does the right thing when nothing is missing, and still sometimes stays silent when something is*. A perfect gate (using Adam's labels as the gate) would score 3.25 — the "oracle" ceiling. We beat the caption baseline (+0.28). The earlier version (v2, one week older) lost clearly under every reference; the week's changes closed the gap to a tie.

A second view of the same result (cost-sensitivity). Because the 4-to-1 lopsidedness comes from the scoring rubric and not from deaf viewers, we also computed a simpler, judge-free score: per clip, did the gate open when a picture was due and stay shut when not? Ours: 39 hits, 11 misses, 28 redundant. Blind: 43 hits, 7 misses, 39 redundant. If a redundant picture is treated as costing a quarter of a miss (the judge's implicit rate), the two tie; if it costs 40% or more, ours is ahead. So: *the gate is preferred as soon as a redundant picture is counted as more than about 40% as bad as a missing one*. Whether real viewers feel it that way is an open question — no user study.

Where the remaining errors come from (read from saved decisions, no new runs):

- Of the 11 test clips where we stayed silent and should not have: **7 are the detector** — the ambient sound was under loud speech or music and never reached the 0.35 bar (a "masked miss"); 4 are debatable labels (cars visible on screen, traffic noise labelled off-screen, like Figure 2b); 0 are timing.
- Of the redundant pictures on the development clips, **about 80% are "phantoms"** — sounds the detector reports confidently that are not there at all (a whale at a Christmas market, roaring lions at a helicopter landing). The gate is right that nothing visible makes them; the picture is a detector mistake.

So the **detector** is the bottleneck on both sides, more than the visibility decision.

Outside check — DCASE 2025. A public research dataset of 30,000 five-second indoor clips in which humans marked, for every sound, when it starts and whether its source is inside the camera's view. We use it as ground truth we did not make ourselves: on 258 of its sounds our visibility decision agrees 66.7% — it keeps 97% of off-screen sounds but silences only 35% of on-screen ones (it errs on the safe side: better a redundant picture than a missing one).

10. What was tried against the detector, and why it stopped

Each attempt had a pass/fail rule written down before it ran, and was reviewed by an independent AI reviewer ("Fable"). None passed; all are reported honestly.

attempt	idea	result
threshold sweep	try 16 gate settings on the development clips	the current setting was already best
CLAP second opinion	a second audio model (CLAP) must agree with BEATs	CLAP recognises only 50% of real sounds at the loosest useful setting — unusable
PANNs agreement	a second AudioSet detector must agree	removed 24 of 77 phantoms (needed 40); both detectors share the same confusions
persistence + rank rule	count a sound only if it stays among the top-3 for ≥ 1 s, instead of one peak	same trade-off curve as the current rule, just shifted
bigger VLM (32B) for the gate	four times larger vision model	recognises visible sources better (67% vs 35%) but wrongly silences 25% of off-screen sounds (was 3%) — worse under the score
audio language model	ask Qwen2-Audio "which of these sounds do you hear?"	small test passed (6/10 phantoms vetoed, 9/10 real kept); full run removed only 25 of 77
source separation (Demucs)	remove speech before detecting	assessed, not run: removes speech only, and 4 of the 7 masked misses are under music

Example of a phantom the second opinions could not remove: "whale vocalization" at a Christmas market (confidence 0.36). The same phantom appears on other crowd clips (a crowd of people, an aquarium walk), so the

detector seems to hear crowd murmur as whale song. The second detector (PANNs) makes the same confusion, so asking it to agree does not help; and a confidence bar high enough to remove the whale also removes real quiet sounds elsewhere.

Conclusion for the thesis: with today's off-the-shelf sound detectors, invented sounds and missed quiet sounds are two ends of one dial; no filter on top fixes both.

11. The benchmark (`benchmark/tags.json`)

274 short clips, labelled by Adam into four scenarios: **unseen** (an ambient sound whose source is off screen — a picture is due), **mixed** (some sources visible, some not — a picture is due), **seen** (the source is on screen — no picture due), **no ambient** (speech or music only — no picture due). The labelling rule is the one professional captioners use: "does the picture already tell the viewer this sound is happening?"

Test set: 100 clips, 25 per scenario — every reported score uses these.

Development split: the other 174 clips, used only for choosing settings. (All 25 mixed clips ended up in the test set; this mattered once, §10.)

Label reliability: Adam re-labelled 60 clips blind and agreed with himself on 78% of them (by the usual scale, "moderate" agreement); the disagreement is mostly on the "unseen" clips, which are exactly the ones the gate is judged on. One annotator; a second would make this firmer.

12. Decisions taken, what is open, and how to reproduce the numbers

Decided: keep the occlusion timing refinement (measured to make no difference; v3 was evaluated with it on). Skip the independent reference for v3 (it cannot score the gate). Stop detector experiments (§10). Bibliography verified and two figures added (done 15 Sept).

Open for Adam: read the report; re-render the three demo videos with v3 (the existing demos predate BEATs; ~1 GPU-hour); final proofread.

Future work (in the report): a detector-only benchmark on the labelled clips; a speech-aware detector measured against it; an audio model with an open vocabulary for sounds AudioSet has no name for ("phone alert"); a second annotator; a study with deaf viewers.

Reproduce the headline without a GPU (the per-clip results are already in the repository):

```
python scripts/paired_stats.py v3 v3_grounded      # 3.17 vs 3.16, the
python scripts/cost_sensitivity.py                 # 39/11/28 vs 43/7,
```

Run everything again (GPU cluster, ~10 h): `bash slurm/sync_data.sh --code-only` uploads the code; `sbatch slurm/job_v3.sh` runs the 100 test clips through the system and the two baselines and scores them (resumable: if it stops, resubmit the same job, never start over). Run one video locally: `python main.py <video>`; settings in `config.py`.

Where everything is.

- **Scores:** `benchmark/protocol_results_<tag>.json` (one line per clip and system, with the reference, the description and the judge's reason).
- **Experiment records:** `benchmark/gate_setting.json`, `clap_setting.json`, `panns_agree.json`, `persist_rank_setting.json`, `eval_dcase_visibility_32b.json`, `audio_llm_eval.json`.
- **Cluster** (BIU Slurm, VPN via F5 at `access.biu.ac.il`): jobs are `slurm/*.sh`. A Hugging Face token in `~/.bashrc` on the cluster is needed for the gated models.
- **Documents:** `docs/report/` (thesis), `docs/project_notes.tex` (lab notebook), `docs/failure_catalogue.md`, `docs/plan_robustness.md` (process rules), `LIMITATIONS.md`.

13. Glossary

- **gate** — the rule "show a picture only if the sound's source is not visible". The idea under test.
- **gated / ours / proposed** — our system. **blind** — draw a picture for every sound heard, never look at the video. **caption** — write the sound's name as text.
- **bar (threshold)** — the confidence (0.35) a sound must reach to count as detected.
- **phantom** — a sound the detector reports that is not there (a whale at a market).
- **masked miss** — a real sound the detector does not reach the bar on because speech or music is louder.

- **onset** — the moment a sound starts. **hysteresis** — stamping the onset where the confidence begins to rise, not where it crosses the bar.
- **stretch** — a piece of at most 5 s of one sound; the visibility question is asked per stretch. **vote** — one of the three differently worded visibility questions.
- **dedup** — merging several detector names for one sound source into one picture.
- **scenario** — Adam's label for a clip: unseen / mixed / seen / no ambient.
- **dev (development) split** — the 174 clips used to choose settings. **test set** — the 100 clips every reported score uses.
- **reference (sentence)** — what a hearing viewer would get that a deaf viewer misses; the thing the judge compares against. Three versions: model-derived, human-grounded, independent.
- **judge** — the text model that scores a system's output against the reference, 0–4.
- **oracle** — the score if the gate used Adam's labels directly (3.25); the ceiling.
- **cost-sensitivity / r** — the judge-free view: how bad a redundant picture is, as a fraction of a missing one. The gate wins from $r = 0.40$.
- **DCASE 2025** — a public dataset with human marks for when each sound starts and whether its source is in view; our outside check.
- **VLM** — vision-language model, a model you can show pictures to and question in words.
- **AudioSet** — Google's 527-kind sound vocabulary; BEATs, PANNs and CLAP all learned from it.